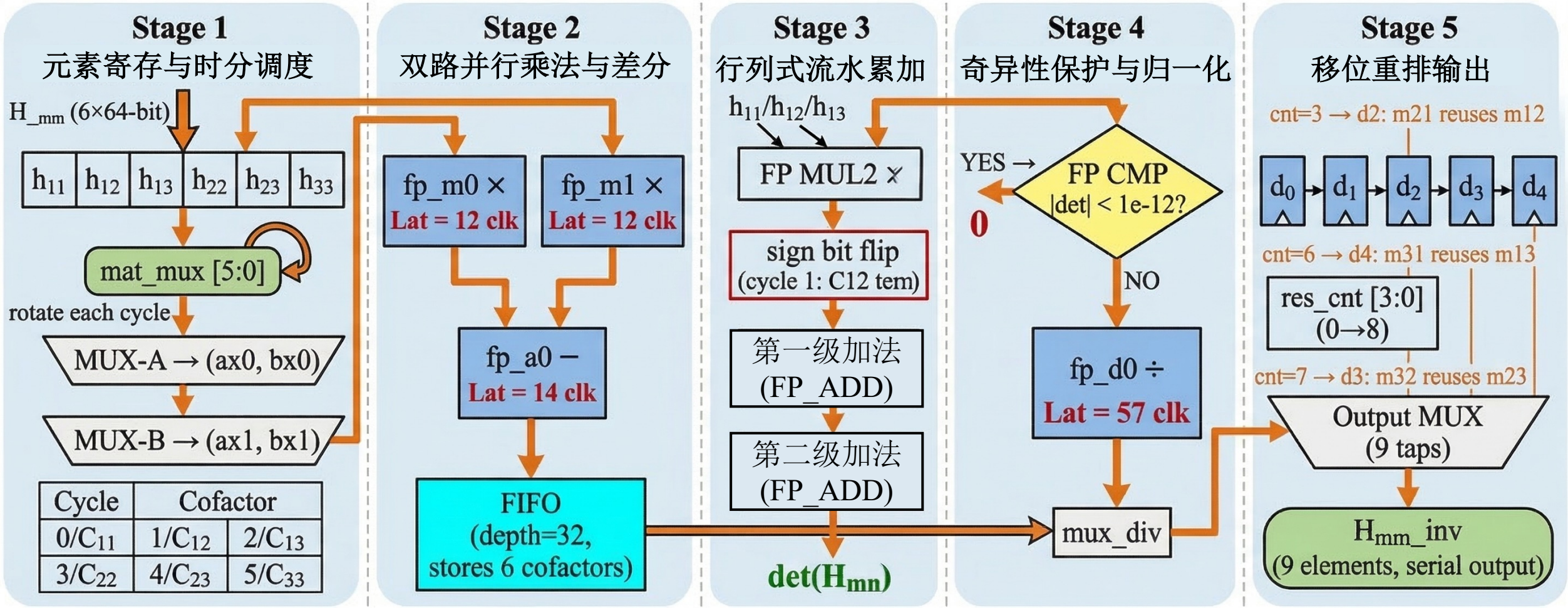
Stage 1
元素寄存与时分调度
H_mm (6×64-bit)
h11 h12 h13 h22 h23 h33
mat_mux [5:0]
rotate each cycle
MUX-A → (ax0, bx0)
MUX-B → (ax1, bx1)
Cycle
Cofactor
0/C11
1/C12
2/C13
3/C22
4/C23
5/C33
Stage 2
双路并行乘法与差分
fp_m0 ×
Lat = 12 clk
fp_m1 ×
Lat = 12 clk
fp_a0 −
Lat = 14 clk
FIFO
(depth=32,
stores 6 cofactors)
Stage 3
行列式流水累加
h11/h12/h13
FP MUL2 ×
sign bit flip
(cycle 1: C12 tem)
第一级加法
(FP_ADD)
第二级加法
(FP_ADD)
det(Hmn)
Stage 4
奇异性保护与归一化
YES →
0
FP CMP
|det| < 1e-12?
NO
fp_d0 ÷
Lat = 57 clk
mux_div
Stage 5
移位重排输出
cnt=3 → d2: m21 reuses m12
d0 d1 d2 d3 d4
cnt=6 → d4: m31 reuses m13
res_cnt [3:0]
(0→8)
cnt=7 → d3: m32 reuses m23
Output MUX
(9 taps)
Hmm_inv
(9 elements, serial output)